

NOGALES, AZ, USA

WMO: 722728

Lat: 31.421N

Lon: 110.846W

Elev: 3932

StdP: 12.72

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 03196

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 26.4	(c) 29.3	(d) -3.4	(e) 5.3	(f) 58.1	(g) 0.9	(h) 6.6	(i) 58.1	(j) 24.4	(k) 55.5	(l) 20.8	(m) 56.6	(n) 2.6	(o) 70	(p) 0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 32.1	(c) 100.0	(d) 62.7	(e) 97.8	(f) 62.5	(g) 95.4	(h) 62.2	(i) 70.5	(j) 83.9	(k) 69.5	(l) 82.7	(m) 68.8	(n) 81.8	(o) 9.8	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.6	117.6	72.9	66.4	112.8	72.7	65.5	109.0	72.6	36.9	84.3	36.1	82.6	35.3	81.5	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.0	(b) 18.4	(c) 16.1	(e) 19.7	(f) 104.9	(g) 3.9	(h) 2.7	(i) 16.9	(j) 106.9	(k) 14.6	(l) 108.5	(m) 12.5	(n) 110.0	(o) 9.7	(p) 112.0		
(a) 21.0	(b) 18.4	(c) 16.1	(e) 16.6	(f) 72.5	(g) 3.6	(h) 1.3	(i) 14.0	(j) 73.5	(k) 11.8	(l) 74.3	(m) 9.8	(n) 75.0	(o) 7.2	(p) 76.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	49.8	51.7	57.1	62.3	69.4	79.7	80.4	78.3	75.0	66.3	57.1	49.3
	DBStd	12.55	5.95	6.56	6.25	5.81	5.63	4.58	3.79	3.57	3.64	5.35	6.34	6.31
	HDD50	258	76	54	17	2	0	0	0	0	0	1	17	91
	HDD65	2010	472	373	250	114	25	0	0	0	0	49	240	487
	CDD50	5645	69	101	237	372	600	892	943	878	749	505	230	69
	CDD65	1924	0	0	5	34	160	442	478	413	299	89	4	0
	CDH74	25299	17	101	415	1114	2977	6582	5014	3989	3196	1554	312	28
	CDH80	12643	0	11	74	324	1389	4005	2718	2014	1516	548	44	0
(14) Wind	WSAvg	5.7	5.0	5.7	6.3	7.0	7.2	6.7	5.2	4.4	5.0	5.4	5.2	5.0
(15) Precipitation	PrecAvg	17.0	1.1	0.9	0.9	0.2	0.2	0.4	4.6	3.5	1.9	1.3	0.7	1.6
	PrecMax	26.9	3.1	2.5	3.8	0.9	0.9	2.2	9.2	7.9	5.2	8.3	3.5	8.0
	PrecMin	9.9	0.0	0.0	0.0	0.0	0.0	0.0	0.7	1.2	0.0	0.0	0.0	0.0
	PrecStd	4.5	0.9	0.8	1.0	0.2	0.3	0.6	2.0	1.8	1.3	1.9	0.8	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.3	81.4	85.8	90.4	98.0	104.4	102.4	99.4	96.0	91.6	84.4	77.4
		MCWB	49.4	50.7	52.9	55.0	59.6	62.5	64.0	66.3	63.6	59.2	53.7	49.9
	2%	DB	72.8	77.2	81.9	86.8	93.6	100.8	99.3	96.7	93.3	88.7	81.0	73.3
		MCWB	47.8	49.5	52.2	53.5	57.0	61.5	64.3	66.4	63.0	58.7	52.9	48.7
	5%	DB	69.9	73.1	78.9	83.8	90.9	98.8	96.7	93.6	91.2	85.9	78.0	69.6
		MCWB	46.7	48.0	50.6	52.5	56.1	61.0	64.4	66.6	63.3	57.7	52.1	47.9
	10%	DB	66.0	68.8	75.4	80.9	88.1	96.6	93.1	90.6	88.9	82.9	74.3	64.5
		MCWB	45.4	46.6	49.0	51.4	55.1	60.3	65.0	66.9	63.1	57.0	50.9	45.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	54.0	56.3	57.1	62.0	68.2	71.7	72.0	69.8	65.3	57.7	53.9
		MCDB	62.7	67.9	78.4	83.8	89.1	86.7	84.1	86.2	83.2	78.1	71.5	64.0
	2%	WB	50.8	51.6	53.4	55.2	59.4	66.2	70.4	70.9	68.5	63.2	55.6	51.2
		MCDB	63.6	68.3	77.0	81.9	86.6	86.5	82.6	85.2	81.9	77.9	71.8	66.7
	5%	WB	48.8	50.1	51.5	53.4	57.7	64.7	69.4	69.9	67.6	61.4	54.0	49.4
		MCDB	64.3	67.7	74.3	79.9	86.4	86.8	81.5	84.1	80.7	77.6	71.6	65.3
	10%	WB	47.0	48.3	50.1	52.0	56.0	63.2	68.7	69.1	66.5	59.4	52.4	47.5
		MCDB	62.9	65.6	73.0	78.1	85.0	88.0	80.9	82.8	79.2	76.6	70.8	62.4
(35) Mean Daily Temperature Range	5% DB	MDBR	29.7	29.1	31.2	32.5	33.0	32.1	24.5	24.7	27.1	30.9	30.5	28.6
		MCDBR	35.9	37.5	37.3	37.1	37.0	34.2	29.7	29.2	30.9	35.0	35.7	35.8
		MCWBR	17.9	17.6	16.8	16.3	15.0	12.5	8.6	8.5	10.2	14.1	16.1	17.6
	5% WB	MCDBR	28.8	30.3	33.2	33.3	33.7	27.9	22.3	23.9	23.9	26.9	28.8	29.6
		MCWBR	15.1	14.8	15.0	14.7	14.2	9.6	6.3	7.3	8.0	10.8	13.5	15.4
(40) Clear-Sky Solar Irradiance	taub		0.255	0.259	0.278	0.287	0.302	0.331	0.386	0.373	0.329	0.293	0.265	0.253
	taud		2.646	2.639	2.554	2.515	2.512	2.437	2.417	2.442	2.538	2.581	2.623	2.646
	Ebn at Noon		316	323	320	318	312	301	284	287	299	305	309	311
	Edh at Noon		24	26	31	34	34	36	37	36	31	28	24	23
(44) All-Sky Solar Radiation	RadAvg		1187	1483	1954	2385	2616	2560	2064	2003	1861	1629	1303	1088
	RadStd		78	117	103	77	90	146	138	77	89	100	62	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.58	N/A	N/A	N/A	+0.94	+1.55	N/A	-290	+308	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A

Nomenclature: See separate page